Assignment 05 CODE -

SERVER -

```
%%writefile server.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 45678
#define BUFFER_SIZE 1024

int main()
{
    int server_socket, client_socket;
    struct sockaddr_in server_address, client_address;
    socklen_t client_length;

    char buffer[BUFFER_SIZE];

    // 1. Create TCP socket
    server_socket = socket(AF_INET, SOCK_STREAM, 0);

    if (server_socket < 0)
    {
        perror("Socket creation failed");
        return 1;
    }

    printf("Server socket created successfully.\n");

    // 2. Configure server address
    server_address.sin_family = AF_INET;
    server_address.sin_addr.s_addr = INADDR_ANY;
    server_address.sin_port = htons(PORT);

    // 3. Bind socket
    if (bind(server_socket,
             (struct sockaddr *)&server_address,
             sizeof(server_address)) < 0)
    {
        perror("Bind failed");
        close(server_socket);
        return 1;
```

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}

printf("Socketboundtoport%d.\n", PORT);

// 4. Listen for client
if (listen(server_socket, 5) < 0)
{
   perror("Listen failed");
   close(server_socket);
   return 1;
}

printf("Server is listening...\n");

// 5. Accept client connection
client_length=sizeof(client_address);

client_socket=accept(server_socket,
                     (structsockaddr*)&client_address,
                     &client_length);

if (client_socket < 0)
{
   perror("Accept failed");
   close(server_socket);
   return 1;
}

printf("Clientconnectedsuccessfully.\n");

//6.Receivemessagefromclient
memset(buffer,0,BUFFER_SIZE);

recv(client_socket,buffer,BUFFER_SIZE - 1, 0);

// 7. Display received message
printf("Messagereceivedfromclient: %s\n", buffer);

// 8. Send reply to client
charreply[]="HelloClient!Message received successfully.";

send(client_socket,reply,strlen(reply), 0);

printf("Reply sent to client.\n");

// 9. Close client connection
close(client_socket);
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    // 10. Close server socket
    close(server_socket);

    printf("Server closed.\n");

    return 0;
}
```

Client -

```
%%writefile client.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

#define PORT 45678
#define BUFFER_SIZE 1024

int main()
{
    int client_socket;
    struct sockaddr_in server_address;

    char message[BUFFER_SIZE];
    char buffer[BUFFER_SIZE];

    // 1. Create TCP socket
    client_socket = socket(AF_INET, SOCK_STREAM, 0);

    if (client_socket < 0)
    {
        perror("Socket creation failed");
        return 1;
    }

    printf("Client socket created successfully.\n");

    // 2. Configure server address
    server_address.sin_family = AF_INET;
    server_address.sin_port = htons(PORT);

    // Server is running on localhost
    if (inet_pton(AF_INET, "127.0.0.1",
                  &server_address.sin_addr) <= 0)
```

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    {
        perror("Invalidaddress");
        close(client_socket);
        return 1;
    }

    // 3. Connect to server
    if (connect(client_socket,
                (structsockaddr*)&server_address,
                sizeof(server_address)) < 0)
    {
        perror("Connectionfailed");
        close(client_socket);
        return 1;
    }

    printf("Connectedtoserver.\n");

    //4.Inputmessagefromuser
    printf("Entermessageforserver: ");
    fgets(message,BUFFER_SIZE, stdin);

    // Remove newline
    message[strcspn(message, "\n")] = '\0';

    //5.Sendmessagetoserver
    send(client_socket,message, strlen(message), 0);

    printf("Messagesenttoserver.\n");

    //6.Receivereplyfromserver
    memset(buffer,0,BUFFER_SIZE);

    recv(client_socket,buffer,BUFFER_SIZE - 1, 0);

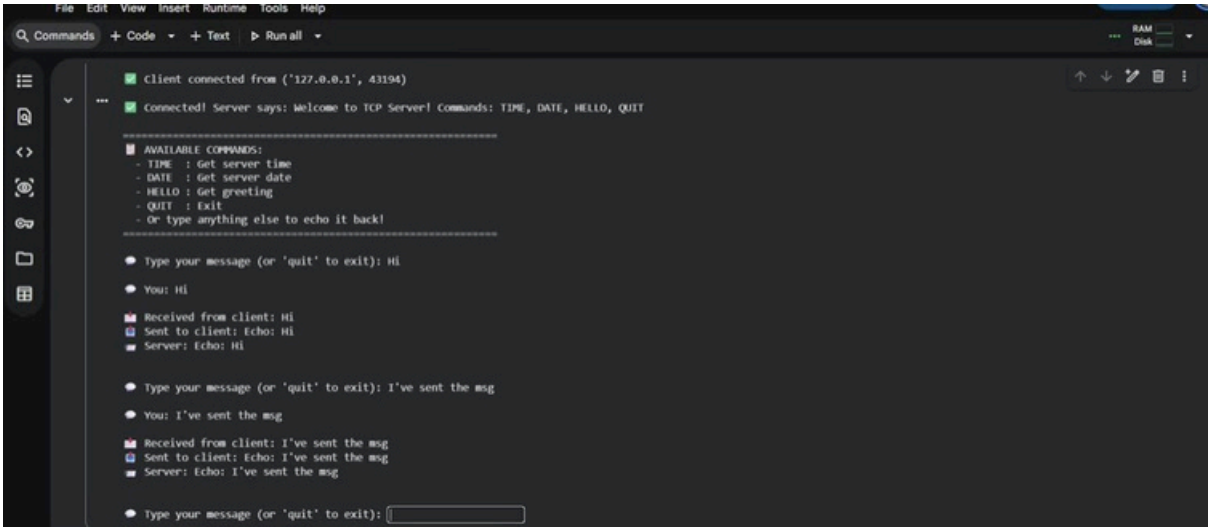
    // 7. Display server reply
    printf("Replyfromserver:%s\n", buffer);

    // 8. Close client socket
    close(client_socket);

    printf("Client closed.\n");

    return 0;
}
```

OUTPUT -



GITHUB LINK-
https://github.com/akkishansh/computer-networks-lab/